Standard paper-and-pencil neuropsychology tests - Insight46 phase 1

Kirsty Lu | 23/07/2019

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Document details

Categories of variables	Standard paper-and-pencil neuropsychology tests
Summary of work undertaken	1. Data collected as described below. 2. FNAME data cleaned as described below. 3. Preclinical Alzheimer's Cognitive Composite (PACC) generated according to methods described below.
Names of source variables	The, following, variables, relate, to, the, MMSE, and, were, all, extracted, from, XNAT, I, have, renamed, them, to, give, them, more, intuitive, names, I, have, provided, an, annotated, copy, of, the, XNAT, MMSE, form, at, the, end, of, this, document, in, order, to, identify, each, variable, id, ID, number), Ecomp, Ewn, Eyear, Eseason, Edate, Eday, Emonth, Ecountry, Ecity, Earea, Efloor, Ename, Eorientation, out, of, for, the, orientation, questions), Ereg_first1, Ereg_second1, Ereg_third1, Ereg_first2, Ereg_second2, Ereg_third2, Ereg_first3, Ereg_second3, Ereg_third3, Esubtract1, Esubtract2, Esubtract3, Esubtract4, Esubtract5, Esubtract_tot, Ebackwards, Ehighest, Erecall_first, Erecall_second, Erecall_third, Ewatch, Epen, Erepetition, Erighthand, Efolded, Elap, Ereading, Esentence, Edrawing, Es, The, following, variables, relate, to, the, Logical, Memory, Matrix, Reasoning, Digit-Symbol, Substitution, and, FNAME, tests, They, were, all, extracted, from, XNAT, I, have, renamed, them, to, give, them, more, intuitive, names, and, provided, annotated, copies, of, the, XNAT, forms, at, the, end, of, this, document, in, order, identify, each, variable, Licomp, Liwn, Lis, Ldcomp, Ldwn, Lt, Lds, Mcomp, Mwn, Ms, Wcomp, Wwn, Ws, Fcomp, Fwn, Finr1, Fior1, Finr2, Fior2, Finrt, Fiort, Fcrn, Fcro, Frecog, Fcrn30, Fcro30, Fmcn, Fmco, Ffnn, Ffno, Ftot
Output variables	mmse_comp_i46p1, mmse_wn_i46p1, mmse_year_i46p1, mmse_season_i46p1, mmse_date_i46p1, mmse_day_i46p1, mmse_month_i46p1, mmse_country_i46p1, mmse_city_i46p1, mmse_area_i46p1, mmse_floor_i46p1, mmse_name_i46p1, mmse_orientation_i46p1, mmse_reg_first1_i46p1, mmse_reg_second1_i46p1, mmse_reg_third1_i46p1, mmse_reg_first2_i46p1, mmse_reg_second2_i46p1, mmse_reg_third2_i46p1, mmse_subtract1_i46p1, mmse_subtract2_i46p1, mmse_subtract3_i46p1, mmse_subtract4_i46p1, mmse_subtract5_i46p1, mmse_subtract_tot_i46p1, mmse_backwards_i46p1, mmse_highest_i46p1, mmse_recall_first_i46p1, mmse_recall_second_i46p1, mmse_recall_third_i46p1, mmse_watch_i46p1, mmse_pen_i46p1, mmse_repetition_i46p1, mmse_righthand_i46p1, mmse_folded_i46p1, mmse_lap_i46p1, mmse_reading_i46p1, mmse_sentence_i46p1, mmse_drawing_i46p1, mmse_total_i46p1, mmse_zscore_i46p1, logmemimmcomp_i46p1, logmemimmtotal_i46p1, logmemimmzscore_i46p1, matrixcomp_i46p1, matrixtotal_i46p1, matrixzscore_i46p1, digsymsubcomp_i46p1, digsymsubnotcomp_i46p1, digsymsubtotal_i46p1, digsymsubzscore_i46p1, logmemdelcomp_i46p1, logmemdeltime_i46p1, logmemdeltotal_i46p1, logmemdelzscore_i46p1, fname_comp_i46p1, fname_wn_i46p1, fname_inr1_i46p1, fname_ior1_i46p1, fname_inr2_i46p1, fname_ior2_i46p1, fname_inrtotal_i46p1, fname_iortotal_i46p1, fname_crn_i46p1, fname_cro_i46p1, fname_recog_i46p1, fname_crn30_i46p1, fname_cro30_i46p1, fname_mcn_i46p1, fname_mco_i46p1, fname_namestotal_i46p1, fname_occstotal_i46p1, fname_total_i46p1, fname_nameszscore_i46p1, fname_occszscore_i46p1, fname_totalzscore_i46p1, pacc_i46p1

Papers using these variables	1. Lu et al. (2019) Cognition at age 70: life course predictors and associations with brain pathologies, Neurology, in press 2. Characterising cognition in later life and its relationship with biomarkers of Alzheimer's disease: a study of members of the Nationa Survey of Health and Development (the British 1946 birth cohort). PhD thesis (Kirsty Lu) 3. Blood and cerebrospinal fluid biomarkers in Alzheimer's disease – from clinical to preclinical cohorts. PhD thesis (Ashvini Keshavan). 4. Plasma amyloid, tau and serum neurofilament light chain in Insight 46 – associations with cognition and brain imaging. AAIC 2019 poster (Keshavan et al.)

Background to cognitive tests

Full details of these cognitive tests have been published in the following two papers:

1. Lane et al. (2017) Study protocol: Insight 46 – a neuroscience sub-study of the MRC National Survey of Health and Development, BMC Neurology, 17:75, doi: 10.1186/s12883-017-0846-x
2. Lu et al. (2019) Cognition at age 70: life course predictors and associations with brain pathologies, Neurology, in press

A summary is provided below:

The **Mini Mental State Examination (MMSE)** is a 30-point composite screening tool for cognitive impairment which is widely used within clinical practice (Folstein, M. F., Folstein, S. E. and McHugh, P. R. (1975) "Mini-mental state." A practical method for grading the cognitive state of patients for the clinician., Journal of Psychiatric Research, 12(3), pp. 189–198. doi: 10.1016/0022-3956(75)90026-6). The item-by-item scores are provided in this dataset, as well as the total score.

Digit-Symbol Substitution from the Wechsler Adult Intelligence Scale-Revised is an index of executive function and psychomotor speed (Wechsler D. (1981b) Wechsler Adult Intelligence Scale - Revised. San Antonio, TX: The Psychological Corporation). The score is the number of items completed correctly within 90 seconds.

Logical Memory IIa from the Wechsler Memory Scale assesses free recall of a short story, which the participant is asked to recall immediately and after a delay of approximately 20 minutes (Wechsler D. (1987) Wechsler Memory Scale - Revised: Manual. San Antonio, TX: Psychological Corporation). The maximum score is 25. The exact delay duration is recorded so that it can be accounted for in analyses.

The **Face-Name test (FNAME-12)** assesses associative memory for face-name and face-occupation pairs (Papp, K. V et al. (2014) Development of a psychometrically equivalent short form of the Face-Name Associative Memory Exam for use along the early Alzheimer's disease trajectory, Clin Neuropsychol. 2014/05/13, 28(5), pp. 771–785. doi:10.1080/13854046.2014.911351). Two versions exist: FNAME-12A and FNAME12-B. This study used FNAME-12A. Participants are shown 12 unfamiliar face-name and face-occupation pairs (e.g. "Sarah, Reporter"), with 8 seconds to study each one. They are then presented with each face and asked to recall the associated name and occupation. This process is repeated with a second learning phase and a second recall test. After a ~10-minute delay they are again shown each face and asked to recall the names and occupations (the third recall test). After a ~30-minute delay participants are shown 12 sets of three faces and asked to identify each previously learned face from the two distractors (facial recognition) and to recall the name and occupation (the fourth recall test). If they cannot recall the name and/or occupation, they are asked to select the correct answer from three options comprising: the correct answer, a distractor (a name/occupation that belongs with a different face in the set), and a name/occupation that did not feature in the set. The summary outcomes are FN-N (total names recalled, max. 48), FN-O (total occupations recalled, max. 48) and FNAME-total (FN-N + FN-O, max. 96) – these outcomes are based on the four recall tests.

The **Matrix Reasoning** test assesses non-verbal reasoning, an aspect of fluid intelligence (Wechsler D. (1999) The Wechsler abbreviated scale of intelligence. San Antonio, TX: The Psychological Corporation). Participants are shown a matrix of geometric shapes and are required to select the missing piece from five options. There are 32 matrices, graded in difficulty, and the test is discontinued when participants make four consecutive errors

(or four errors within five consecutive items), as specified in the manual. Note that the full test contains 35 items but the final 3 items are only to be administered to younger adults, as per the instructions in the manual, so were not used in Insight 46.

Data cleaning

One participant did not complete the full FNAME test, and therefore her total scores (Ffnn, Ffno, Ftot) automatically generated in XNAT were not valid. These total scores were changed to missing. A note to explain this was added to the Fwn variable for this participant.
On the MMSE, It was noted that 2 participants who failed to name all 3 items correctly on the first attempt (see question 11 in the MMSE form below) had missing data for their second attempt. This omission does not affect their MMSE score, since the score for the naming items is based solely on the first attempt. This is likely to be an administration error, as other versions of the MMSE do not require multiple attempts on this question. No other data-cleaning was required.

Calculation of the Preclinical Alzheimer Cognitive Composite (PACC)

The four components of our version of the PACC were: MMSE total score, Logical Memory delayed recall score, Digit-Symbol Substitution score and FNAME-total. The four components were converted into z-scores based on the full Insight 46 sample, using the following Stata command:

egen newvar = std(oldvar)

The four z-scores were then averaged. A higher PACC score indicates better performance. Two participants did not complete the FNAME test and one did not complete the Digit-Symbol Substitution test. For these three participants, their PACC score was the average of the z-scores for the three tests they completed. Note that the original PACC (Donohue, M. C. et al. (2014) 'The preclinical Alzheimer cognitive composite: measuring amyloid-related decline.', JAMA neurology, 71(8), pp. 961–70. doi: 10.1001/jamaneurol.2014.803) contained the Free and Cued Selective Reminding Test (FCSRT) instead of FNAME. We replaced the FCSRT with FNAME to avoid potential overlap with a similar word-learning memory test administered to the NSHD cohort at multiple time-points throughout adulthood.

MMSE XNAT form - annotated in red with names of variables contained in this dataset

May I ask you some questions about your memory? Ecomp	0 No		
	1 Yes		
2 Unable for health reasons			
3 Unable, other _ _ _ _ _ _ _ _ _ _ _ _			
If 0, 2 or 3 are ticked, specify reason Ewn			
ORIENTATION			
1	What is the year? _ _ _ _ Eyear	0 Incorrect	1 Correct
2	What is the season? _ _ _ _ _ _ Eseason	0 Incorrect	1 Correct
3	What is the date? _ _/ _ _/ _ _ _ _ (DD/MM/YYY) Edate	0 Incorrect	1 Correct
4	What is the day of the week? _ _ _ _ _ Eday	0 Incorrect	1 Correct

5	What is the month? _ _ _ _ _ _ Emonth	0 Incorrect	1 Correct
6	Can you tell me where you are now? For instance what country are we in? _ _ _ _ _ _ Ecountry	0 Incorrect	1 Correct
7	What is the name of this city? _ _ _ _ _ _ Ecity	0 Incorrect	1 Correct
8	What area of the city are we in? Earea	0 Incorrect	1 Correct
9	What floor of the building are we on? _ _ _ _ _ _ _ Efloor	0 Incorrect	1 Correct
10	What is the name of this place? _ _ _ _ _ _ _ _ Ename	0 Incorrect	1 Correct
REGISTRATION			
11	I am now going to name three objects. The three objects are: APPLE, TABLE, PENNY. Please repeat the name of these three objects back to me. Record the FIRST responses (order of object recall does NOT matter):		
First object named? _ _ _ _ _ _Ereg_first1	0 Incorrect	1 Correct	
Second object named? _ _ _ _ _ _ Ereg_second1	0 Incorrect	1 Correct	
Third object named? _ _ _ _ _ _ Ereg_third1	0 Incorrect	1 Correct	
	If all three objects are repeated correctly (in any order) go to Q12a If the participant does not repeat all three words exactly then allow them two further attempts but DO NOT change their first responses or scores. Attempt 2: I'm going to repeat once more the three objects, APPLE, TABLE, PENNY. Please say them back to me. Ereg_first2 Ereg_second2 Ereg_third2 Attempt 3: Can we try one more time? The names are, APPLE, TABLE, PENNY. Please say them back to me. Ereg_first3 Ereg_second3 Ereg_third3		

ATTENTION AND CALCULATION			
12a	Please take seven away from one hundred (Answer: 93) Answer 1: _ _ Esubtract1	0 Incorrect	1 Correct
Now I'm going to ask you to take seven away from what you have left over, and then keep taking seven away until I stop you. (Answers: 86, 79, 72 and 65. If the participant gets an answer wrong but then takes seven away correctly from that answer you should score as correct). Answer 2: _ _ Esubtract2	0 Incorrect	1 Correct	
Answer 3: _ _ Esubtract3	0 Incorrect	1 Correct	
Answer 4: _ _ Esubtract4	0 Incorrect	1 Correct	
Answer 5: _ _ Esubtract5	0 Incorrect	1 Correct	
	If the participant scores 5/5 then go to Q13. If the participant scores less than 5/5 for 12a then complete question 12b.	Total score: _ /5 Esubtract_tot	
12b	Please spell "WORLD" backwards (Answer: D L R O W)	TOTAL number of letters in correct order _/5 Ebackwards	
	Highest score for subtraction or backwards spelling tests	_ /5 Ehighest (note that this variable is only entered for participants who were administered item 12b)	
RECALL			
13	Please name the three objects that I mentioned to you earlier?		
First object named _ _ _ _ _ _ _ Erecall_first	0 Incorrect	1 Correct	
Second object named _ _ _ _ _ _ _ Erecall_second	0 Incorrect	1 Correct	
Third object named _ _ _ _ _ _ _ Erecall_third	0 Incorrect	1 Correct	
NAMING			
14	What is this called? (show watch) _ _ _ _ _ _ Ewatch What is this called? (show pen) _ _ _ _ _ _ Epen	0 Incorrect	1 Correct
15	0 Incorrect	1 Correct	
REPETITION			

16	I am going to say a sentence to you and I would like you to repeat it after I have said it. The sentence is: “NO IFS, ANDS OR BUTS.” Did the participant answer correctly? Do not repeat the sentence. If necessary say: I’m sorry, I’m only allowed to say the sentence once. Erepetition	0 Incorrect	1 Correct
THREE STAGE COMMAND			
17	Please take this paper with your right hand, fold it in half and put it on your lap. Do not repeat the sentence. If necessary say: I’m sorry, I’m only allowed to read that out once. Did the participant take the paper in their right hand? Erighthand	0 No	1 Yes
Did the participant fold the paper in half or quarter (both allowed)? Efolded	0 No	1 Yes	
Did the participant put the paper on their lap? Elap	0 No	1 Yes	
READING			
18	Now give the sheet called “MMSE CLOSE YOUR EYES” to the participant and say: Please read the sentence at the top of this sheet and do what it says. Did the participant close their eyes? Ereading	0 No	1 Yes
WRITING			
19	Please write a sentence in the space here. Indicate the space under “Write a sentence”. Did the participant write a sentence? Esentence	0 No	1 Yes
20	Please copy this drawing. Indicate the space to the right of the design. Did the participant copy the drawing correctly? Edrawing	0 No	1 Yes
Score	_ _ /30 Es		

Logical Memory Immediate Recall XNAT form - annotated in red with names of variables contained in this dataset

WMS Logical Memory – Immediate Recall completed Licomp	0 No

1 Yes	
2 Unable for health reasons	
3 Unable, other (Specify)_ _ _ _ _ _ _ _	
If 0, 2 or 3 are ticked, specify reason Liwn	
Total number of story units recalled:	_ _ /25 Lis

Logical Memory Delayed Recall XNAT form - annotated in red with names of variables contained in this dataset

WMS Logical Memory – Delayed completed Ldcomp	0 No
1 Yes	
2 Unable for health reasons	
3 Unable, other (Specify)_ _ _ _ _ _ _ _ Ldwn	
If 0, 2 or 3 are ticked, specify reason Ldwn	
Total number of story units recalled:	_ _ /25 Lds
Time elapsed since Logical Memory-Immediate:	_ _ (minutes) Lt

Matrix Reasoning XNAT form - annotated in red with names of variables contained in this dataset

WASI Matrix Reasoning completed Mcomp	0 No
1 Yes	
2 Unable for health reasons	
3 Unable, other (Specify)_ _ _ _ _ _ _ _ Mwn	
If 0, 2 or 3 are ticked, specify reason Mwn	
Total correct:	_ _ /32 Ms

Digit-Symbol XNAT form - annotated in red with names of variables contained in this dataset

WAIS Digit Symbol Substitution Test completed Wcomp	0 No
1 Yes	
2 Unable for health reasons	
3 Unable, other (Specify)_ _ _ _ _ _ _ _ Wwn	
If 0, 2 or 3 are ticked, specify reason Wwn	
Total correct:	_ _ /93 Ws

FNAME XNAT form - annotated in red with names of variables contained in this dataset

Face Name Associative Memory Exam completed Fcomp	0 No
1 Yes	
2 Unable for health reasons	
3 Unable, other (Specify)_ _ _ _ _ _ _ _ Fwn	
If 0, 2 or 3 are ticked, specify reason Fwn	
Learning Trial 1	
INR1: Finr1 IOR1: Fior1	_ _ /12 _ _ /12
Learning Trial 2	

INR2: Finr2 IOR2: Fior2	_ _ /12 _ _ /12
Total INR: Finrt Total IOR: Fiort	_ _ /24 _ _ /24

Cued Name and Occupation Recall	
CRN: Fcrn CRO: Fcro	_ _ /12 _ _ /12

Delayed Cued Recall of Names and Occupations	
Facial Recognition Frecog	_ _ /12
CRN 30: Fcrn30 CRO30: Fcro30	_ _ /12 _ _ /12
MCN: Fmcn MCO: Fmco	_ _ /12 _ _ /12
Totals	
FN-N (Total INR + CRN + CRN30) Ffnn	_ _ /48
FN-O (Total IOR + CRO + CRO30 Ffno	_ _ /48
Total FNAME Score (CN-N + FN-O) Ftot	_ _ /96

Variables in this document

74 high-confidence matches

Variable	Label	How it was found
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Associative episodic memory [60429]		
fname_comp_i46p1	Face-Name test (FNAME): was it completed? - Insight46 phase 1	topsheet field
fname_crn30_i46p1	Face-Name test (FNAME): CRN30 Cued Name Retrieval long delay - Insight46 phase 1	topsheet field
fname_crn_i46p1	Face-Name test (FNAME): CRN Cued Recall Names short delay - Insight46 phase 1	topsheet field
fname_cro30_i46p1	Face-Name test (FNAME): CRO30 Cued Occupation Retrieval long delay - Insight46 phase 1	topsheet field
fname_cro_i46p1	Face-Name test (FNAME): CRO Cued Recall Occupations short delay - Insight46 phase 1	topsheet field
fname_inr1_i46p1	Face-Name test (FNAME): INR1 Names Learning Trial 1 - Insight46 phase 1	topsheet field
fname_inr2_i46p1	Face-Name test (FNAME): INR2 Names Learning Trial 2 - Insight46 phase 1	topsheet field
fname_inrtotal_i46p1	Face-Name test (FNAME): INR Total Names Learning Trials 1 and 2 - Insight46 phase 1	topsheet field
fname_ior1_i46p1	Face-Name test (FNAME): IOR1 Occupations Learning Trial 1 - Insight46 phase 1	topsheet field
fname_ior2_i46p1	Face-Name test (FNAME): IOR2 Occupations Learning Trial 2 - Insight46 phase 1	topsheet field
fname_iortotal_i46p1	Face-Name test (FNAME): IOR Total Occupations Learning Trials 1 and 2 - Insight46 phase 1	topsheet field
fname_mcn_i46p1	Face-Name test (FNAME): MCN Multiple Choice Names - Insight46 phase 1	topsheet field
fname_mco_i46p1	Face-Name test (FNAME): MCN Multiple Choice Occupations - Insight46 phase 1	topsheet field

Variable	Label	How it was found
fname_namestotal_i46p1	Face-Name test (FNAME): FN-N Names Total Score - Insight46 phase 1	topsheet field
fname_nameszscore_i46p1	Face-Name test (FNAME): Z-score of FN-N total names score - Insight46 phase 1	topsheet field
fname_occstotal_i46p1	Face-Name test (FNAME): FN-O Occupations Total Score - Insight46 phase 1	topsheet field
fname_occszscore_i46p1	Face-Name test (FNAME): Z-score of FN-O total occupations score - Insight46 phase 1	topsheet field
fname_recog_i46p1	Face-Name test (FNAME): Facial Recognition - Insight46 phase 1	topsheet field
fname_total_i46p1	Face-Name test (FNAME): Total Score - Insight46 phase 1	topsheet field
fname_totalzscore_i46p1	Face-Name test (FNAME): Z-score of total score - Insight46 phase 1	topsheet field
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Fluid intelligence and reasoning [60424]		
matrixcomp_i46p1	Matrix Reasoning: was it completed? - Insight46 phase 1	topsheet field
matrixtotal_i46p1	Matrix Reasoning: score - Insight46 phase 1	topsheet field
matrixzscore_i46p1	Matrix Reasoning: Z-score - Insight46 phase 1	topsheet field
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Logical memory [60430]		
logmemdelcomp_i46p1	WMS-R Logical Memory Delayed: was it completed? - Insight46 phase 1	topsheet field
logmemdeltime_i46p1	WMS-R Logical Memory: time elapsed for delay (min) - Insight46 phase 1	topsheet field
logmemdeltotal_i46p1	WMS-R Logical Memory Delayed: score - Insight46 phase 1	topsheet field
logmemdelzscore_i46p1	WMS-R Logical Memory Delayed: Z-score - Insight46 phase 1	topsheet field
logmemimmcomp_i46p1	WMS-R Logical Memory Immediate: was it completed? - Insight46 phase 1	topsheet field
logmemimmtotal_i46p1	WMS-R Logical Memory Immediate: score - Insight46 phase 1	topsheet field
logmemimmzscore_i46p1	WMS-R Logical Memory Immediate: Z-score - Insight46 phase 1	topsheet field
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Mini-mental state examination [60422]		
mmse_area_i46p1	MMSE: orientation - what area of the city are we in? - Insight46 phase 1	topsheet field
mmse_backwards_i46p1	MMSE: attention and calculation - WORLD backwards (out of 5), administered if subtraction not perfect - Insight46 phase 1	topsheet field
mmse_city_i46p1	MMSE: orientation - what is the name of this city? - Insight46 phase 1	topsheet field
mmse_comp_i46p1	MMSE: May I ask you some questions about your memory? - Insight46 phase 1	topsheet field
mmse_country_i46p1	MMSE: orientation - Can you tell me where you are now, for instance what country are we in? - Insight46 phase 1	topsheet field
mmse_date_i46p1	MMSE: orientation - what is the date - Insight46 phase 1	topsheet field
mmse_day_i46p1	MMSE: orientation - what is the day of the week - Insight46 phase 1	topsheet field
mmse_drawing_i46p1	MMSE: copy drawing of overlapping pentagons - Insight46 phase 1	topsheet field

Variable	Label	How it was found
mmse_floor_i46p1	MMSE: orientation - what floor of the building are we on? - Insight46 phase 1	topsheet field
mmse_folded_i46p1	MMSE: command - fold paper - Insight46 phase 1	topsheet field
mmse_highest_i46p1	MMSE: attention and calculation - highest out of subtraction or WORLD backwards if both were administered (out of 5) - Insight46 phase 1	topsheet field
mmse_lap_i46p1	MMSE: command - put paper on lap - Insight46 phase 1	topsheet field
mmse_month_i46p1	MMSE: orientation - what is the month - Insight46 phase 1	topsheet field
mmse_name_i46p1	MMSE: orientation - what is the name of this place - Insight46 phase 1	topsheet field
mmse_orientation_i46p1	MMSE: orientation total out of 10 - Insight46 phase 1	topsheet field
mmse_pen_i46p1	MMSE: naming - pen - Insight46 phase 1	topsheet field
mmse_reading_i46p1	MMSE: reading (close your eyes) - Insight46 phase 1	topsheet field
mmse_recall_first_i46p1	MMSE: recall - first object - Insight46 phase 1	topsheet field
mmse_recall_second_i46p1	MMSE: recall - second object - Insight46 phase 1	topsheet field
mmse_recall_third_i46p1	MMSE: recall - third object - Insight46 phase 1	topsheet field
mmse_reg_first1_i46p1	MMSE: registration - first object attempt 1 - Insight46 phase 1	topsheet field
mmse_reg_first2_i46p1	MMSE: registration - first object attempt 2 - Insight46 phase 1	topsheet field
mmse_reg_second1_i46p1	MMSE: registration - second object attempt 1 - Insight46 phase 1	topsheet field
mmse_reg_second2_i46p1	MMSE: registration - second object attempt 2 - Insight46 phase 1	topsheet field
mmse_reg_third1_i46p1	MMSE: registration - third object attempt 1 - Insight46 phase 1	topsheet field
mmse_reg_third2_i46p1	MMSE: registration - third object attempt 2 - Insight46 phase 1	topsheet field
mmse_repetition_i46p1	MMSE: repetition - Insight46 phase 1	topsheet field
mmse_righthand_i46p1	MMSE: command - take paper in right hand - Insight46 phase 1	topsheet field
mmse_season_i46p1	MMSE: orientation - what is the season - Insight46 phase 1	topsheet field
mmse_sentence_i46p1	MMSE: write a sentence - Insight46 phase 1	topsheet field
mmse_subtract1_i46p1	MMSE: attention and calculation - subtraction 1 - Insight46 phase 1	topsheet field
mmse_subtract2_i46p1	MMSE: attention and calculation - subtraction 2 - Insight46 phase 1	topsheet field
mmse_subtract3_i46p1	MMSE: attention and calculation - subtraction 3 - Insight46 phase 1	topsheet field

Variable	Label	How it was found
mmse_subtract4_i46p1	MMSE: attention and calculation - subtraction 4 - Insight46 phase 1	topsheet field
mmse_subtract5_i46p1	MMSE: attention and calculation - subtraction 5 - Insight46 phase 1	topsheet field
mmse_subtract_tot_i46p1	MMSE: attention and calculation - subtraction total (out of 5) - Insight46 phase 1	topsheet field
mmse_total_i46p1	MMSE: score - Insight46 phase 1	topsheet field
mmse_watch_i46p1	MMSE: naming - watch - Insight46 phase 1	topsheet field
mmse_year_i46p1	MMSE: orientation - what is the year - Insight46 phase 1	topsheet field
mmse_zscore_i46p1	MMSE: Z-score - Insight46 phase 1	topsheet field
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Preclinical Alzheimer Cognitive Composite (PACC) [60432]		
pacc_i46p1	Preclinical Alzheimer Cognitive Composite (PACC) - Insight46 phase 1	topsheet field
Sub-Studies [57] › Insight46 [699] › Cognitive function [6042] › Symbol digit substitution [60425]		
digsymsubcomp_i46p1	WAIS Digit Symbol Substitution Test: was it completed? - Insight46 phase 1	topsheet field
digsymsubtotal_i46p1	WAIS Digit Symbol Substitution Test: score - Insight46 phase 1	topsheet field
digsymsubzscore_i46p1	WAIS Digit Symbol Substitution Test: Z-score - Insight46 phase 1	topsheet field

1 medium-confidence match

Variable	Label	How it was found
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Other medical conditions [51011]		
blood	Have you had anaemia in the last 10 years - at age 43 years	prose scan

2 low-confidence matches

Variable	Label	How it was found
Clinical data [51] › Physical measures [110] › Hand examination [1106]		
hand	Now I would like to examine your hands for any bumps or swellings. Would you be willing for me to examine your hands? - first record - at age 53 years	prose scan
Questionnaire data [55] › Health and medical history [510] › Medical conditions [5101] › Musculoskeletal system conditions [51016]		
back	In the last 12 months, have you had sciatica, lumbago or severe backache at age 53 years	prose scan